INCH-POUND

MS51821F
24 May 2018
SUPERSEDING
MS51821E
30 July 1987

DETAIL SPECIFICATION SHEET

# TEE, PIPE TO TUBE, FEMALE PIPE ON BRANCH, FLARELESS TYPE, HYDRAULIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Reactivated after 24 May 2018 and may be used for new and existing designs and acquisitions.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.

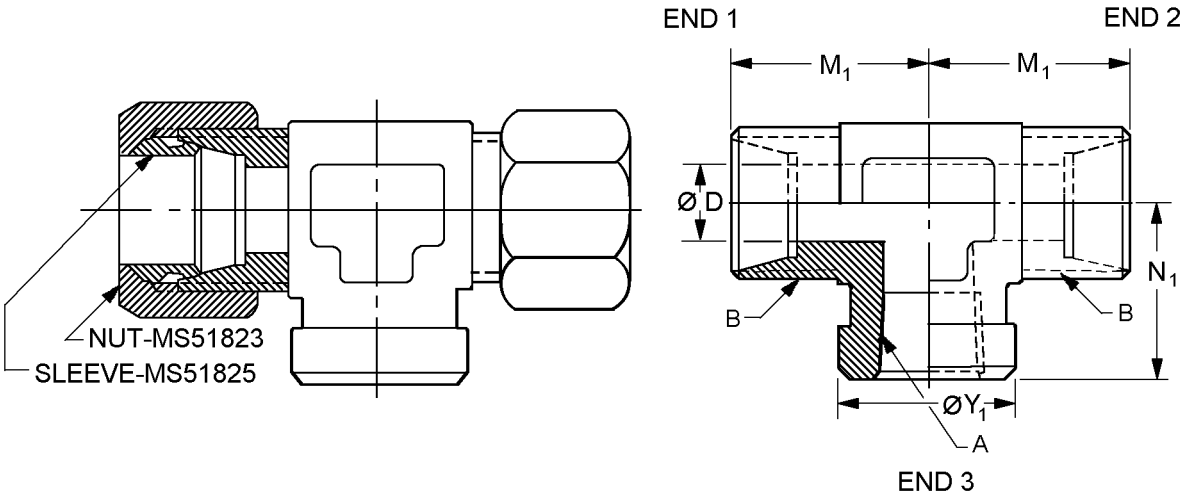



FIGURE 1. Tee, pipe to tube.

AMSC N/A

FSC 4730

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| Dash number (see note 5) | | Tube OD Nom inches (mm) | | A Thread NPTF | B Straight thread | D diameter Drill |
|---|---|---|---|---|---|---|
| Assembly | Body | End 1 | End 2 | End 3 | | |
| -1 | -21 | .1250 (3.175) | | .125 -27 | .3125 - 24UNF | .093 (2.36) |
| -2 | -22 | .1875 (4.763) | | .125 -27 | .3750 - 24UNF | .125 (3.18) |
| -3 | -23 | .2500 (6.350) | | .125 -27 | .4375 - 20UNF | .203 (5.16) |
| -4 | -24 | .3125 (7.936) | | .125 -27 | .5000 - 20UNF | .234 (5.94) |
| -5 | -25 | .3750 (9.525) | | .250 - 18 | .5625 - 18UNF | .281 (7.14) |
| -6 | -26 | .5000 (12.700) | | .375 - 18 | .7500 - 16UNF | .422 (10.72) |
| -7 | -27 | .6250 (15.875) | | .500 - 14 | .8750 - 14UNF | .500 (12.70) |
| -8 | -28 | .7500 (19.050) | | .750 - 14 | 1.0625 - 12UN | .656 (16.66) |
| -9 | -29 | .8750 (22.225) | | .750 - 14 | 1.1875 - 12UN | .718 (18.24) |
| -10 | -30 | 1.0000 (25.400) | | 1.000 - 11 1/2 | 1.3125 - 12UN | .875 (22.23) |
| -11 | -31 | 1.2500 (31.750) | | 1.250 - 11 1/2 | 1.6250 - 12UN | 1.093 (27.76) |
| -12 | -32 | 1.5000 (38.100) | | 1.500 - 11 1/2 | 1.8750 - 12UN | 1.344 (34.14) |
| -13 | -33 | 2.0000 (50.800) | | 2.000 - 11 1/2 | 2.5000 - 12UN | 1.813 (46.05) |

| Dash number (see note 5) | | $M_1$ inches (mm) ±.020 (0.51) | $N_1$ inch mm) ±.03 (0.8) | $Y_1$ inch (mm) ±.020 (0.51) |
|---|---|---|---|---|
| Assembly | Body | | | |
| -1 | -21 | .83 (21.08) | .66 (16.8) | .562 (14.27) |
| -2 | -22 | .83 (21.08) | .66 (16.8) | .562 (14.27) |
| -3 | -23 | .89 (22.61) | .66 (16.8) | .562 (14.27) |
| -4 | -24 | .95 (24.13) | .66 (16.8) | .562 (14.27) |
| -5 | -25 | 1.05 (26.67) | .88 (22.4) | .750 (19.05) |
| -6 | -26 | 1.23 (31.24) | 1.02 (26.0) | .875 (22.23) |
| -7 | -27 | 1.42 (36.07) | 1.23 (31.2) | 1.062 (26.97) |
| -8 | -28 | 1.58 (40.13) | 1.36 (34.5) | 1.312 (33.32) |
| -9 | -29 | 1.62 (41.15) | 1.42 (36.1) | 1.312 (33.32) |
| -10 | -30 | 1.73 (43.94) | 1.62 (41.2) | 1.625 (41.28) |
| -11 | -31 | 2.08 (52.83) | 1.70 (43.2) | 1.875 (47.63) |
| -12 | -32 | 2.58 (65.53) | 2.08 (52.9) | 2.562 (65.07) |
| -13 | -33 | 2.64 (67.06) | 2.39 (60.7) | 2.812 (71.42) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514.
5. Assemblies are only furnished to this specification (body, nut, and sleeve). Bodies are not to be stocked, stored, or issued.
6. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.

FIGURE 1. Tee, pipe to tube - Continued.



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REQUIREMENTS:

Tee, pipe to tube shall be as specified on figure 1 and in table I.

Materials shall be in accordance with MIL-DTL-18866.

Finish. Finishes shall be as specified in table I. All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.



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TABLE I. Material and finish identification codes.

| PIN code material/plating finish | Material | Plating finish |
|---|---|---|
| Blank | Steel | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/ |
| H | Steel | Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 2/ |
| J | Steel | Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/ 3/ |
| M | Nickel-copper alloy UNS N04400 | No additional finish. |
| N | High-chromium nickel alloy UNS N06690 | No additional finish. |
| P | Steel | Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 4/ |
| R | Steel | Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 4/ 5/ |
| SS | Corrosion resistant steel | No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7, free machining alloy UNS S30300 method 1, type 2 or 4. |
| T | Titanium | Anodize in accordance with SAE-AMS2488 type 2. 6/ |
| TF | Titanium | Fluoride phosphate in accordance with SAE-AMS2486. 6/ |
| Z | Steel | Zinc plating in accordance with ASTM B695, type II, class 8. 4/ 5/ |
| ZC | Steel | Zinc may be any zinc plating from PIN codes J, R, and Z with a colored chromate coating. 4/ 5/ |

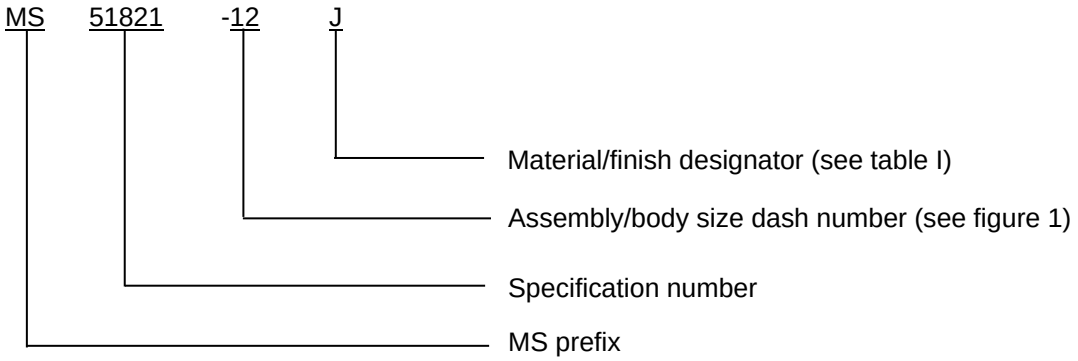
1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA owned shipboard systems.
2/ Hexavalent chromium free. Finish shall be RoHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
4/ Not for use in aircraft. Requires approval from the Program Office for all applications. Zinc-plated materials should be avoided in hydraulic systems with water-glycol fluids. Reference ASTM MNL37 - Fuels and Lubricants Handbook, "water glycol fluids are highly alkaline due to the presence of amine based corrosion inhibitors. As a result, these fluids can attack zinc, cadmium, magnesium, and non-anodized aluminum, forming sticky or gummy residues. Consequently, these metals should be avoided when selecting system components.
5/ These finishes have a chromate conversion coating which includes hexavalent chromium and are not recommended.
6/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293. Titanium shall not be used in oxygen or potable water systems.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.



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PIN: The PIN consists of the letters “MS”, the specification number, a dash, number for a tee pipe to tube assembly/body size, and a letter for material finish designator.




PIN example: MS51821-12J indicates a tee, pipe to tube, female pipe on branch assembly, .6250 inch (15.875 mm) tube OD on each branch, to 1/2 NPTF, steel with zinc-nickel finish.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2.

Class I and II ozone depleting substances (ODS) shall not be used in MS51821 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Table II provides a detail reference data listing for MS39208 to MS51821. This table contains coordination of equivalent tube sizes, in both compression and hydraulic type tee’s. Assembly replacement is authorized.

TABLE II. Assembly equivalency (for information only).

| PIN | | PIN | |
|---|---|---|---|
| Compression MS39208 | Hydraulic MS51821 | Compression MS39208 | Hydraulic MS51821 |
| -1 | -1 | -7 | -7 |
| -2 | -2 | -8 | -5 |
| -3 | -3 | -9 | -6 |
| -4 | -4 | -10 | -7 |
| -5 | -5 | -11 | -8 |
| -6 | -6 | -12 | -12 |



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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

| | | |
|---|---|---|
| MIL-DTL-16232 | SAE-AMS-STD-595/36076 | SAE-AMS-STD-595/36270 |
| MS39208 | SAE-AMS-STD-595/36081 | SAE-AMS-STD-595/36280 |
| MS51823 | SAE-AMS-STD-595/36099 | SAE-AMS-STD-595/36293 |
| MS51825 | SAE-AMS-STD-595/36118 | SAE-AMS2417 |
| ASTM B117 | SAE-AMS-STD-595/36134 | SAE-AMS2486 |
| ASTM B633 | SAE-AMS-STD-595/36152 | SAE-AMS2488 |
| ASTM B695 | SAE-AMS-STD-595/36170 | SAE-AMS2700 |
| ASTM F1136/F1136M | SAE-AMS-STD-595/36173 | SAE-J514 |
| ASTM MNL37 | SAE-AMS-STD-595/36176 | T9070-AL-DPC-020/077-2 |
| SAE-AMS-C-81562 | SAE-AMS-STD-595/36231 | |
| SAE-AMS-QQ-P-416 | SAE-AMS-STD-595/36251 | |

## CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - OS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2018-057)

Review activities:
Army - AT, GL, MI
Navy - AS, MC
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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